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CREATE SCHEMA IF NOT EXISTS finalproject;
SET search_path TO finalproject;
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DROP TABLE IF EXISTS ReturnRefund CASCADE;
DROP TABLE IF EXISTS Payment CASCADE;
DROP TABLE IF EXISTS Shipment CASCADE;
DROP TABLE IF EXISTS OrderItem CASCADE;
DROP TABLE IF EXISTS "Order" CASCADE;
DROP TABLE IF EXISTS CartItem CASCADE;
DROP TABLE IF EXISTS Cart CASCADE;
DROP TABLE IF EXISTS WishListItem CASCADE;
DROP TABLE IF EXISTS WishList CASCADE;
DROP TABLE IF EXISTS Review CASCADE;
DROP TABLE IF EXISTS Coupon CASCADE;
DROP TABLE IF EXISTS Address CASCADE;
DROP TABLE IF EXISTS Customer CASCADE;
DROP TABLE IF EXISTS PurchaseOrder CASCADE;
DROP TABLE IF EXISTS Inventory CASCADE;
DROP TABLE IF EXISTS VendorProduct CASCADE;
DROP TABLE IF EXISTS ProductVariant CASCADE;
DROP TABLE IF EXISTS Product CASCADE;
DROP TABLE IF EXISTS Category CASCADE;
DROP TABLE IF EXISTS Vendor CASCADE;
DROP TABLE IF EXISTS Warehouse CASCADE;
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CREATE TABLE Category (
    CategoryID SERIAL PRIMARY KEY,
    CategoryName VARCHAR(100) NOT NULL,
    ParentCatID INT,
    FOREIGN KEY (ParentCatID) REFERENCES Category(CategoryID)
    ON DELETE SET NULL
);
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CREATE TABLE Product (
    ProductID SERIAL PRIMARY KEY,
    ProductName VARCHAR(150) NOT NULL,
    MRP NUMERIC(10,2) NOT NULL,
    Description TEXT,
    Brand VARCHAR(100),
    Weight NUMERIC(8,2),
    SKU VARCHAR(50) UNIQUE NOT NULL,
    CategoryID INT NOT NULL,
    FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)
    ON DELETE CASCADE
);
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CREATE TABLE ProductVariant (
    VariantID SERIAL PRIMARY KEY,
    Color VARCHAR(50),
    Size VARCHAR(20),
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VariantSKU VARCHAR(50) UNIQUE NOT NULL,
PriceOffset NUMERIC(10,2) DEFAULT 0,
ProductID INT NOT NULL,
FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
ON DELETE CASCADE
);

CREATE TABLE Vendor (
VendorID SERIAL PRIMARY KEY,
CompanyName VARCHAR(150) NOT NULL,
Email VARCHAR(100) UNIQUE NOT NULL,
Phone VARCHAR(15) UNIQUE,
GSTNumber VARCHAR(20) UNIQUE NOT NULL,
ContactPerson VARCHAR(100),
Rating NUMERIC(3,2) DEFAULT 0.00,
CHECK (Rating >= 0 AND Rating <= 5)
);

CREATE TABLE VendorProduct (
VP_ID SERIAL PRIMARY KEY,
SellingPrice NUMERIC(10,2) NOT NULL,
IsAvailable BOOLEAN NOT NULL DEFAULT TRUE,
VendorID INT NOT NULL,
VariantID INT NOT NULL,
UNIQUE (VendorID, VariantID),
CHECK (SellingPrice > 0),
FOREIGN KEY (VendorID) REFERENCES Vendor(VendorID)
ON DELETE CASCADE,
FOREIGN KEY (VariantID) REFERENCES ProductVariant(VariantID)
ON DELETE CASCADE
);

CREATE TABLE Warehouse (
WarehouseID SERIAL PRIMARY KEY,
WarehouseName VARCHAR(100) NOT NULL,
City VARCHAR(100) NOT NULL,
State VARCHAR(100) NOT NULL,
PINCode VARCHAR(10) NOT NULL UNIQUE,
Capacity INT NOT NULL,
CHECK (Capacity > 0)
);

CREATE TABLE Inventory (
InventoryID SERIAL PRIMARY KEY,
Quantity INT NOT NULL DEFAULT 0,
OrderLevel INT NOT NULL,
BinLocation VARCHAR(50),
LastUpdated TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
VariantID INT NOT NULL,
WarehouseID INT NOT NULL,

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    CHECK (Quantity >= 0),
    CHECK (OrderLevel >= 0),
    FOREIGN KEY (VariantID) REFERENCES ProductVariant(VariantID)
        ON DELETE CASCADE,
    FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID)
        ON DELETE CASCADE
);

CREATE TABLE PurchaseOrder (
    PO_ID SERIAL PRIMARY KEY,
    OrderDate DATE NOT NULL DEFAULT CURRENT_DATE,
    Status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    ExpDelivDate DATE,
    VendorID INT NOT NULL,
    CHECK (Status IN ('Pending', 'Confirmed', 'Shipped', 'Delivered',
'Cancelled')),
    FOREIGN KEY (VendorID) REFERENCES Vendor(VendorID)
        ON DELETE CASCADE
);

CREATE TABLE PurchaseOrderItem (
    PO_ID INT NOT NULL,
    VP_ID INT NOT NULL,
    Quantity INT NOT NULL,
    PRIMARY KEY (PO_ID, VP_ID),
    CHECK (Quantity > 0),
    FOREIGN KEY (PO_ID) REFERENCES PurchaseOrder(PO_ID)
        ON DELETE CASCADE,
    FOREIGN KEY (VP_ID) REFERENCES VendorProduct(VP_ID)
        ON DELETE CASCADE
);

CREATE TABLE Customer (
    CustomerID SERIAL PRIMARY KEY,
    CustomerName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    Phone VARCHAR(15) UNIQUE NOT NULL,
    PasswordHash VARCHAR(255) NOT NULL,
    CreatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE Address (
    AddressID SERIAL PRIMARY KEY,
    Street VARCHAR(200) NOT NULL,
    City VARCHAR(100) NOT NULL,
    State VARCHAR(100) NOT NULL,
    Pincode VARCHAR(10) NOT NULL
);

CREATE TABLE CustomerAddress (

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    CustomerID INT NOT NULL,
    AddressID INT NOT NULL,
    IsDefault BOOLEAN NOT NULL DEFAULT FALSE,
    PRIMARY KEY (CustomerID, AddressID),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
        ON DELETE CASCADE,
    FOREIGN KEY (AddressID) REFERENCES Address(AddressID)
        ON DELETE CASCADE
);

CREATE TABLE Review (
    ReviewID SERIAL PRIMARY KEY,
    Rating INT NOT NULL,
    Comment TEXT,
    ReviewDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CustomerID INT NOT NULL,
    VP_ID INT NOT NULL,
    CHECK (Rating >= 1 AND Rating <= 5),
    UNIQUE (CustomerID, VP_ID),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
        ON DELETE CASCADE,
    FOREIGN KEY (VP_ID) REFERENCES VendorProduct(VP_ID)
        ON DELETE CASCADE
);

CREATE TABLE Coupon (
    CouponID SERIAL PRIMARY KEY,
    Code VARCHAR(50) UNIQUE NOT NULL,
    DiscountType VARCHAR(20) NOT NULL,
    DiscountValue NUMERIC(10,2) NOT NULL,
    ExpiryDate DATE NOT NULL,
    UsageLimit INT NOT NULL DEFAULT 1,
    CHECK (DiscountType IN ('Percentage', 'Flat')),
    CHECK (DiscountValue > 0),
    CHECK (UsageLimit > 0)
);

CREATE TABLE "Order" (
    OrderID SERIAL PRIMARY KEY,
    OrderDate TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    TotalAmount NUMERIC(10,2) NOT NULL,
    Status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    CustomerID INT,
    AddressID INT,
    CouponID INT,
    CHECK (TotalAmount >= 0),
    CHECK (Status IN ('Pending', 'Confirmed', 'Shipped', 'Delivered',
'Cancelled')),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
        ON DELETE SET NULL,

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        FOREIGN KEY (AddressID) REFERENCES Address(AddressID)
            ON DELETE SET NULL,
        FOREIGN KEY (CouponID) REFERENCES Coupon(CouponID)
            ON DELETE SET NULL
    );

CREATE TABLE OrderItem (
    OrderItemID SERIAL PRIMARY KEY,
    Quantity INT NOT NULL,
    UniPrice NUMERIC(10,2) NOT NULL,
    OrderID INT NOT NULL,
    VP_ID INT NOT NULL,
    CHECK (Quantity > 0),
    CHECK (UniPrice > 0),
    FOREIGN KEY (OrderID) REFERENCES "Order"(OrderID)
        ON DELETE CASCADE,
    FOREIGN KEY (VP_ID) REFERENCES VendorProduct(VP_ID)
        ON DELETE CASCADE
);

CREATE TABLE Payment (
    PaymentID SERIAL PRIMARY KEY,
    GateWayName VARCHAR(100) NOT NULL,
    TxnRef VARCHAR(100) UNIQUE NOT NULL,
    Amount NUMERIC(10,2) NOT NULL,
    Status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    PaidAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    OrderID INT NOT NULL UNIQUE,
    CHECK (Amount > 0),
    CHECK (Status IN ('Pending', 'Success', 'Failed', 'Refunded')),
    FOREIGN KEY (OrderID) REFERENCES "Order"(OrderID)
        ON DELETE CASCADE
);

CREATE TABLE Shipment (
    ShipmentID SERIAL PRIMARY KEY,
    Tracking VARCHAR(100) UNIQUE,
    CourierPartner VARCHAR(100) NOT NULL,
    ETA DATE,
    Status VARCHAR(20) NOT NULL DEFAULT 'Pending',
    DispatchDate DATE,
    OrderID INT NOT NULL UNIQUE,
    WarehouseID INT NOT NULL,
    InventoryID INT,
    CHECK (Status IN ('Pending', 'Dispatched', 'InTransit', 'Delivered',
'Returned')),
    FOREIGN KEY (OrderID) REFERENCES "Order"(OrderID)
        ON DELETE CASCADE,
    FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID)
        ON DELETE RESTRICT,

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        FOREIGN KEY (InventoryID) REFERENCES Inventory(InventoryID)
            ON DELETE SET NULL
    );
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CREATE TABLE Cart (
    CartID SERIAL PRIMARY KEY,
    CustomerID INT NOT NULL UNIQUE,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);
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CREATE TABLE CartItem (
    CartID INT NOT NULL,
    VP_ID INT NOT NULL,
    Quantity INT NOT NULL DEFAULT 1,
    AddedDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (CartID, VP_ID),
    CHECK (Quantity > 0),
    FOREIGN KEY (CartID) REFERENCES Cart(CartID)
        ON DELETE CASCADE,
    FOREIGN KEY (VP_ID) REFERENCES VendorProduct(VP_ID)
        ON DELETE CASCADE
);
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```
CREATE TABLE WishList (
    WishListID SERIAL PRIMARY KEY,
    WishListName VARCHAR(100) NOT NULL DEFAULT 'My WishList',
    CreatedDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);
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CREATE TABLE WishListItem (
    WListItemID SERIAL PRIMARY KEY,
    AddedDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    WishListID INT NOT NULL,
    VariantID INT NOT NULL,
    UNIQUE (WishListID, VariantID),
    FOREIGN KEY (WishListID) REFERENCES WishList(WishListID)
        ON DELETE CASCADE,
    FOREIGN KEY (VariantID) REFERENCES ProductVariant(VariantID)
        ON DELETE CASCADE
);
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CREATE TABLE ReturnRefund (
    ReturnID SERIAL PRIMARY KEY,
    Reason TEXT NOT NULL,
    Status VARCHAR(20) NOT NULL DEFAULT 'Requested',
    RefundAmount NUMERIC(10,2),
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ReqDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
OrderItemID INT NOT NULL,  
CHECK (Status IN ('Requested', 'Approved', 'Rejected', 'Refunded')),  
CHECK (RefundAmount >= 0),  
FOREIGN KEY (OrderItemID) REFERENCES OrderItem(OrderItemID)  
    ON DELETE CASCADE  
);
```